

24/08/2026 -

Model Practical
Examination

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Computer Networks

Configuration of DHCP in Packet Tracer

Aim :

To configure DHCP in Packet Tracer.

Software required :

Packet Tracer / End devices, Hubs, connectors

Algorithm :

1. Start

→ Set up network topology in Packet Tracer with DHCP server and DHCP clients connected to a switch.

2. Configure the DHCP server.

→ Assign the IP address to the server interface.

→ Enable the DHCP service on the server.

3. Configure the switch.

→ Enable the switch interfaces that connect the DHCP clients.

4. Configure the DHCP clients.

5. Client request and server response.

6. client selection and request

The client receives multiple offer messages from different DHCP server if available.

7. server Acknowledgement :

8. configure the client configuration.

9. Lease renewal and expiration.

10. End.

Procedure :

1. Launch cisco Packet tracer and create a new network topology or open an existing one.

2. configure the DHCP server.

3. Then configure the switch.

4. save the configuration.

5. set the IP address to DHCP or obtain an IP address automatically.

6. Save the configuration for each client.

7. start the simulation.

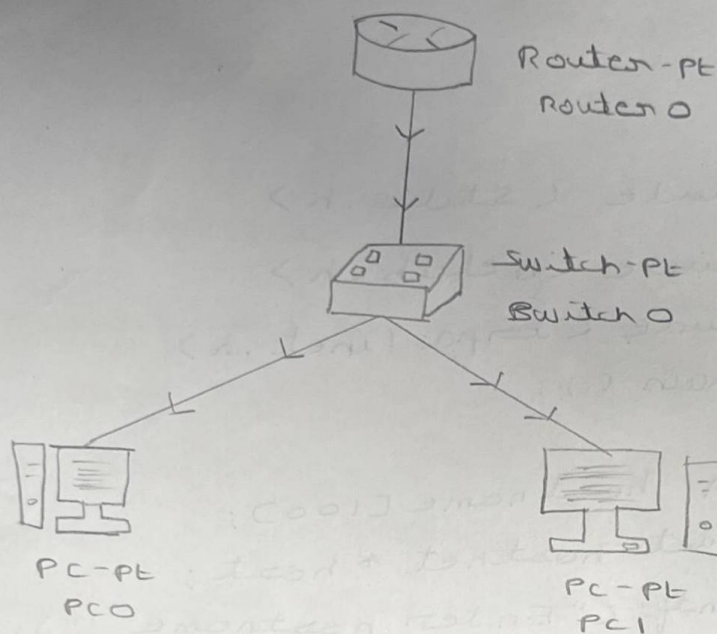
—> click the "start / stop simulation" button to start the simulation.

8. verify DHCP operation.

Result :

Therefore the configuration for DHCP has been successfully executed using packet tracer.

Output :



Time (sec)	Last Device	At Device	Type
0.000	-	PC0	ICMP
0.001	PC0	Switch0	ICMP
0.002	Switch0	PC1	ICMP
0.003	PC1	Switch0	ICMP
0.004	Switch0	PC0	ICMP
1.829	-	Switch0	STP
1.830	Switch0	PC1	STP
1.830	Switch0	PC0	STP

Developing a client that contacts a given DNS Server to resolve a given Host name in Java / C .

Aim :

To develop a client that contacts a given DNS server to resolve a given hostname .

Description :

Get the host name to be resolve using `gethostname()`; check hostname using `nslookup`. Print the IP address, hostname, Address length and address type .

Algorithm :

- step 1 : Find the hostname by using `get host by name()`.
- step 2 : The host name is followed by the list of alias name .
- step 3 : pointer points to the array of pointers to the individual addresses.
- step 4 : For each address call the `inet_ntop()` and Print the returned string .

Procedure :

1. Initialize necessary variables and structure .

2. Send the DNS query to the DNS server.

3. Then receive the DNS response from the DNS server.

→ Use the 'recvfrom()' function to receive the DNS response from the DNS server using the socket descriptor.

→ Store the response in a buffer.

4. Parse and process the DNS response.

5. Display or use the resolved information.

→ Extract the resolved IP address or other relevant data from the DNS response.

→ Process the resolved information as needed in your application.

b. close the socket.

→ use the 'close()' function to close the socket descriptor.

Result :

Thus, the client that connects a given DNS server to resolve a given hostname is developed successfully.

1.5. D HCP

2.8.

2. output :

Code :

```
#include <stdio.h>
#include <netdb.h>
#include <arpa/inet.h>
int main ()
{
    char hostname[100];
    struct hostnet * host;
    printf ("Enter hostname :");
    scanf ("%s", hostname);
    host = gethostbyname (hostname);
    if (host == NULL)
    {
        printf ("Host not found \n");
        return 1;
    }
    printf ("Hostname : %s \n", hostname);
    printf ("IP Address : %s \n",
        inet_ntoa (*(struct in_addr *)
            host->h_addr));
    return 0;
}
```

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